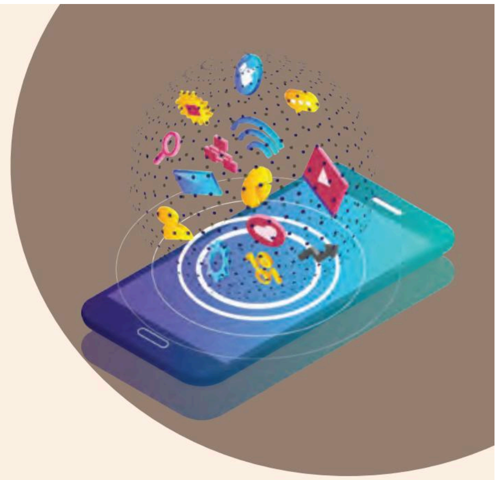
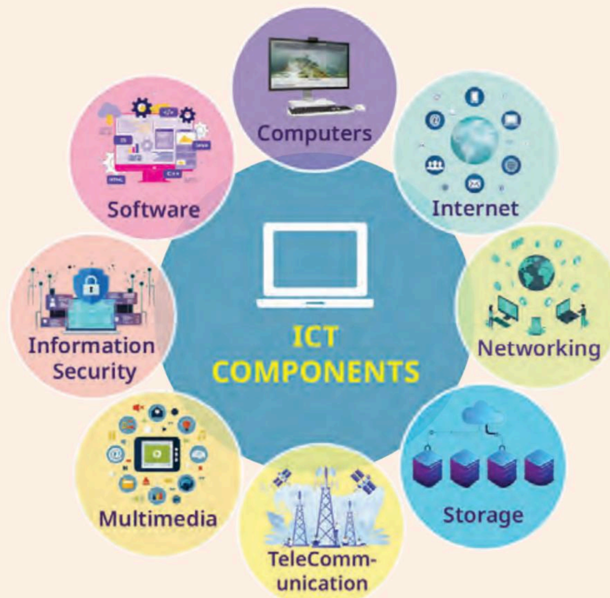


7

Role of ICT

The modern society is greatly driven by Information and Communications Technology. ICT stands for Information and Communication Technology. It is a broad term that includes a wide range of technologies and tools used for handling, processing, storing, and communicating information. All communication devices, cell phones, radio, television, and computers along with satellite systems, are ICT tools.

It is a combination of wired and wireless networking tools that enables receiving and transmitting information and communicating through varied mediums.



Key Components of ICT

The following are the key components of ICT:

Computers Desktops, Personal computers, laptops, tablets, and smartphones help in processing and storing data.

Software Software applications like operating systems, OpenOffice applications (Writer, Calc, Impress, Base), multimedia tools, web browsers, and specialised applications perform specific tasks and help run ICT devices.

Internet The Internet is a global network that connects computers and servers. It allows users to get information, communicate, and share data across the world.

Telecommunication Communication technologies like telephones (landlines and mobiles), messaging, emails, video conferencing, and social media platforms are part of ICT.

Networking Computer networks, both wired and wireless, simplify the exchange of information between devices and across the internet. Wired (e.g., Ethernet) or wireless (e.g., Wi-Fi) networking enables the creation of local area networks (LANs) within homes or businesses and a wide area networks (WANs) that connect computers across larger geographical areas.

Data Storage Hard drives, solid-state drives (pen drives, SD cards), cloud storage (Google drive, iCloud), and external storage devices help ICT in storing data.

Think and Tell

When was the last time you saw or used a landline phone to make a call?

Error Alert!

The Cloud is Actually in the Sky! Cloud, literally means the sky, but in computer language, the cloud refers to remote servers and data centres located all around the world.

Information Security It involves measures and protocols needed for protecting the information and ensuring the confidentiality, integrity, and availability of data and information. It helps prevent data breaches, unauthorised access, and cyberattacks.

Multimedia ICT allows integration of text, images, audio, video, and interactive elements, contributing to richer content experiences.

Think and Tell

You and your friends are making a presentation on a software application. What are the things that you will use to make your presentation interesting?

Role of ICT

ICT tools have become an integral part of our lives and have a huge impact on the way we live, work, and interact with each other.

Importance of ICT in Personal Life

ICT has deeply impacted our personal lives. It has made possible the convenience, efficiency, connectivity, and accessibility of information and resources at every doorstep. It has become part of our lives at varied levels.

Purpose	Tools
Connecting with people	Smartphones, Messaging Apps, Social Media Platforms.
Entertainment	Television, Radio, Tablets, Computer Applications, and Web Browsers for watching movies, listening to music and news.
Learning and information	Internet enables us to watch educational videos, to find answers to questions, learn new things online.
Online Shopping	Applications that allow people to shop easily from the comfort of their homes.
Organisation	Tools like note-taking and digital calendars enhance the productivity of the people.
Money management	Online banking apps have made sending and receiving of money secure and convenient. It can be used for paying bills.
Making friends	Social media platforms have helped people connect with those who share common interests and make new friends.
Healthy living	Many health and fitness-related apps have allowed people to keep track of their daily steps and oxygen levels.
Travel	ICT has enabled people to travel across the globe by easy booking of train/bus/airplane tickets and hotels.

Importance of ICT at Workplace

ICT plays a very significant role in our workplace and has been contributing immensely to the economy through its seamless integration of varied sectors like education, business, communication, entertainment, agriculture, and research. At the workplace, it contributes to improved efficiency through different computer software and applications. These enable keeping important data safe and organised; collaborating with others on projects; training and learning skills needed for performing the job effectively; and presenting data and information using multimedia elements.

Example: In the education sector, the integration of ICT tools has enhanced the learning experiences through the use of videos, audios, images, and online quizzes while making learning more interactive. It provides a wide range of content for the students and teachers to learn from. They enable teachers to record students' data and map their learning journey.

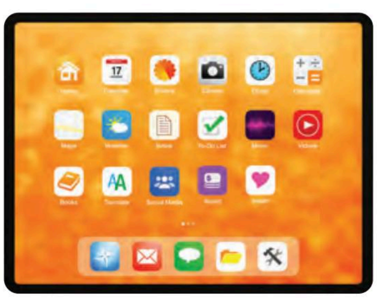
Sectors	Use of ICT
Healthcare	It helps in maintaining electronic health records, telemedicine, testing, and consultancy for patients living far away.
Banking and finance	It has enabled online banking, making payments through mobile apps, and managing finances efficiently.
Agriculture	It helps in weather forecasting and enables farmers to do research and training in the field of agriculture for better farming practices.
Manufacturing and industry	ICT can help automate the production process and make use of the latest technologies and machinery for mass production and manufacturing.
Business	Tools like note-taking and digital calendars enhance the productivity of the people.
Advertising and marketing	It enables marketing and advertising for products and services on online platforms through social media campaigns and analytics for advertising agencies.
Fashion and retail	It has helped in the online selling of goods and services to customers through applications, allowing virtual try-ons, and supply chain management for the fashion industry.

We are highly dependent on these tools and technology for our day-to-day life and for the efficient and effective functioning of our work. The constant development and integration of these technologies have transformed the way we live, work, learn, and interact, thereby improving our quality of life.

ICT Tools

ICT tools like smartphones, tablets, radio, televisions, laptops, and computers are part and parcel of modern society. The accessibility of these tools has made our daily chores easier and improved our quality of life.

Smartphones Mobile phones are portable communication devices that allow people to make and receive calls and send and receive messages. On the other hand, smartphones are advanced mobile phones. They allow one to do things one normally does on a computer. It has a touchscreen that permits users to browse the internet, take pictures, play games, and use multiple apps with touch. It has many helpful features like note-taking, voice assistance, GPS for maps, setting reminders, and so on. They have multiple apps for communications, entertainment, shopping, and banking purposes.



Tablets Tablets are portable computing devices that are bigger than smartphones but smaller than laptops. They have a flat, rectangular design with a touchscreen interface. Unlike laptops, they do not have a physical keyboard. Tablets are lightweight, easy to carry, and have a wide range of functions. They also have productivity applications like word processors, spreadsheets, and presentation software, making them useful for work-related tasks. They are suitable for a wide range of activities, making them popular among individuals of all ages for both personal and professional use.

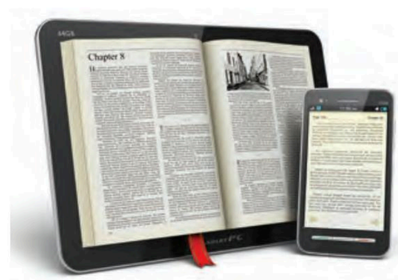
Did You Know?

Google's original name was "Backrub." It was changed to "Google" in 1997, inspired by the word "googol," which refers to the number 1 followed by 100 zeros, representing the vast amount of information the search engine aimed to organise.

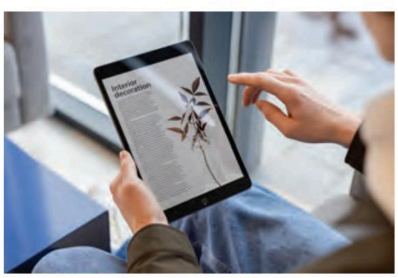
Television Televisions, also known as TVs, are being used for a long time. They were invented in 1927, and since then, they have been making a significant impact on our society by influencing people, shaping culture, entertaining, and giving information. Television uses moving pictures with sounds. It has brought entertainment, news, and educational shows inside every house. Televisions have evolved over the years, and now we also have smart TVs that can connect to the internet and provide a wide range of options from online sources.

Radio Radio is one of the oldest and first ICT tools. It allows people to listen to audio broadcasts for entertainment, information, weather forecasts, commentary, and educational purposes. Radio uses radio waves to broadcast audio in long range. With the advent of the internet, radio has become a more interactive medium, allowing users to access content from anywhere and at any time, at their convenience.

Email Electronic Mail, commonly known as email, enables the exchange of digital messages between individuals and organisations. It has become a primary method of communication in both personal and professional environments. One can have access to their email on their smartphones, laptops, and tablets. It requires an internet connection and allows users to attach files in the form of images, audio, video, or documents.



E-newspaper An e-newspaper, or electronic newspaper, is a digital version of a traditional print newspaper that is accessible through electronic devices such as computers, tablets, smartphones, and e-readers. This allows readers to access news content anytime and anywhere, whether they're commuting, travelling, or simply relaxing at home. E-newspapers can provide real-time updates and live coverage through text updates, images, and even live video streams.



E-readers An e-reader, short for electronic reader or e-book reader, is a specialised electronic device designed primarily for reading digital books and other written content. It serves as an ICT tool with a focus on helping readers enjoy the experience of reading at their own convenience. This is an environment-friendly initiative, allowing readers to have access to a library of books at their fingertips.

Difference Between ICT Tools

Aspects	Smartphones	Tablets	Television	Radio
Type	Portable audio-visual device	Portable audio-visual device	Audio-visual device	Audio broadcast
Communication type	Two-way	Two-way	One-way	One-way
Screen	Small touchscreen, that can fit in pocket	Bigger than smartphones, touchscreen	Larger than smartphones and tablets, may or may not be touchscreen	No screen

Portability (Easy to carry)	Yes	Yes	No	Yes
Applications	Uses many applications	Uses many applications	May or may not use some applications	No applications used
Purpose	Communication, Entertainment, Information Access	Communication, Entertainment, Information Access	Entertainment, Information Access	Audio Content for Communication, Entertainment, Information Access
Internet	Yes	Yes	May or May not SMART TV uses internet	May or May not Podcasts are forms of radio broadcasts that use internet
Storage	Comes with different storage capacity	Comes with higher storage capacity	No storage capacity	No storage capacity
Battery	Operates on rechargeable battery	Operates on rechargeable battery	Operates on power, not battery	Operates on power, not battery
Economical	Comes in all price ranges	Expensive than smartphones	Comes in all price ranges	Cheapest of all four
Accessibility over large distance	Can be accessed from places with network and internet coverage	Can be accessed from places with network and internet coverage	Accessible only within a given room or area	Can be accessed through satellite or internet coverage
Content control/ moderation	No control	No control	Controlled by Censor Board	Controlled by Telecom Regulartory Authority
GPS	Yes	Yes	No	No

Activity Time

Activity 1: ICT in Routine Life *(Group Work)*

In a group of 4–5 students, perform a role play presenting the important role of ICT tools in our personal and work lives. Each role-play should include at least two scenes showing the personal scenario and a work environment.

Activity 2: Importance of ICT *(Group Work)*

Divide the class into small groups of 4–5 students. Ask the groups to choose one of the two topics and discuss the role of ICT. They shall make a poster/ collage on the selected topic based on the ideas discussed.

Topics:

Role of ICT at Workplace

Role of ICT in personal life

Activity 3: Advertising ICT Tools *(Group Work)*

In a group of 4–5 students, prepare a sales pitch for one of the ICT tools. Highlight the key advantages of the chosen tool in your presentation. You may prepare a poster to support your presentation.

Chapter Checkup

A **Select the correct option.**

1. What is the main benefit of using ICT for information access?
   - a. Limited access to information
   - b. Instant access to a wide range of information
   - c. Offline access only
   - d. Access to information only during the daytime
2. Which aspect of personal life can be enhanced by ICT through online courses and tutorials?
   - a. Physical fitness
   - b. Social interactions
   - c. Language learning
   - d. Shopping experience
3. What is the significance of ICT in emergency situations?
   - a. No relevance in emergencies
   - b. Provides only entertainment during emergencies
   - c. Offers access to emergency services, information, and resources
   - d. Causes more problems during emergencies

B **Fill in the blanks with the most suitable words.**

1. The use of ICT tools has led to the rapid exchange of information and .... across the globe.
2. ICT tools have improved .... in the workplace, allowing teams to work together efficiently.
3. One of the potential risks of using ICT tools is .... theft, where sensitive information is accessed without authorisation.
4. .... is an ICT tool that enables users to search for information on the World Wide Web.

C **State whether the following is *True* or *False*. Correct the statements that are false.**

1. E-readers are electronic devices designed for playing games and watching movies.
2. ICT has no role in managing personal finances or budgeting.
3. Remote work and online learning are made possible by ICT tools like video conferencing and collaboration platforms.
4. ICT primarily focuses on communication and does not impact other areas of life.

D **Answer the following questions. *(Solved)***

Q1. Explain the meaning of "ICT tools" and provide two examples of commonly used ICT tools.

A1. ICT tools also known as Information and Communication Technology, is a broad term for various technological devices, software applications, and platforms that help us to communicate, access information, process data, and perform various tasks in personal, professional, educational, and other sectors.

Two examples of commonly used ICT tools are:

Smartphones: Smartphones are versatile ICT tools that combine communication, computing, and connectivity features. They allow users to make calls, send messages, access the internet, use apps for various purposes, take photos and videos, and perform a wide range of tasks.

Email Services: Email services like Gmail, Outlook, and Yahoo Mail provide platforms for sending and receiving electronic messages over the Internet. They enable communication through text, attachments, and media files, making it easy to correspond professionally and personally.

Q2. Describe how ICT tools have revolutionised the healthcare sector.

A2. ICT tools have sparked a profound revolution in the healthcare sector, transforming how medical services are delivered, accessed, and managed. Telemedicine platforms and video conferencing enable remote consultations, connecting patients with healthcare professionals, regardless of location. Electronic Health Records (EHRs) digitise patient information, streamlining data management, reducing errors, and improving care coordination. Smart watches and mobile apps empower patients to monitor vital signs and chronic conditions in real-time, allowing for proactive interventions.

Q3. Differentiate smartphones and tablets on the basis of any three aspects.

A3. Smartphones and tablets are different from each other in the following ways:

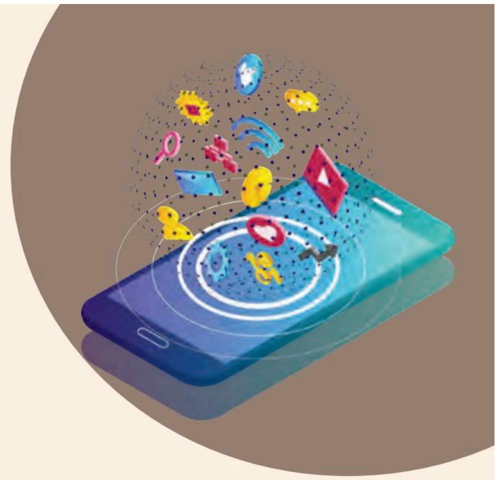
Aspects	Smartphones	Tablets	Television	Radio
Purpose	Communication, Entertainment, Information Access	Communication, Entertainment, Information Access	Entertainment, Information Access	Audio Content for Communication, Entertainment, Information Access
Economical	Comes in all price ranges	Expensive than smartphones	Comes in all price ranges	Cheapest of all four
Storage	Comes with different storage capacity	Comes with higher storage capacity	No storage capacity	No storage capacity

Answer Key

A 1. b 2. c 3. c

B 1. communication 2. efficiency 3. cyber/data 4. Internet

C 1. False. E-readers are electronic devices designed for reading digital books and other written content.
2. False. ICT has a significant role in managing personal finances or budgeting.
3. True.
4. False. ICT has a huge impact on the way we live, work, and interact with each other.



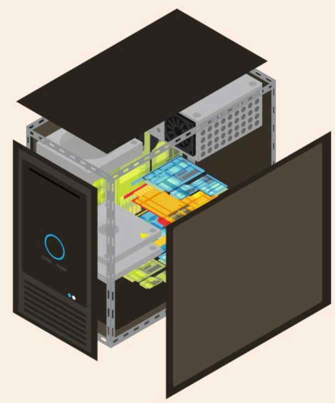
8

Components of Computer System

A computer system comprises various interdependent components that work collectively to execute tasks and process data. Understanding these components is crucial to understanding the functioning of a computer system.

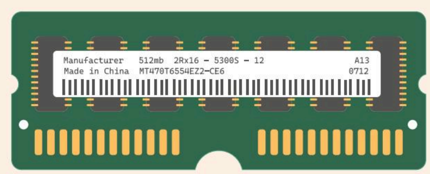
Basic Components of a Computer

Following are the basic components of a computer system:



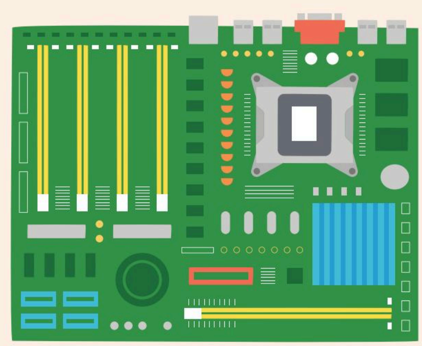
Central Processing Unit (CPU)

The Central Processing Unit (CPU) is known as the brain of the computer. It executes instructions and performs calculations. It consists of the Arithmetic Logic Unit (ALU), the Control Unit (CU), and the Memory Unit (MU). The ALU performs arithmetic and logical operations, while the Control Unit coordinates and manages the various components of the CPU.



Memory

Memory in a computer system is vital for storing and accessing data quickly. There are two types of memory: Random Access Memory (RAM) and Read-Only Memory (ROM). RAM is a type of computer memory that is used to store data and the machine code that is currently being used. It is a volatile memory, which means that data is stored temporarily in RAM and lost forever when the computer is turned off. On the other hand, ROM is a type of memory from which information can only be read. It is a non-volatile memory, as data is stored permanently in ROM and cannot be altered.



Motherboard

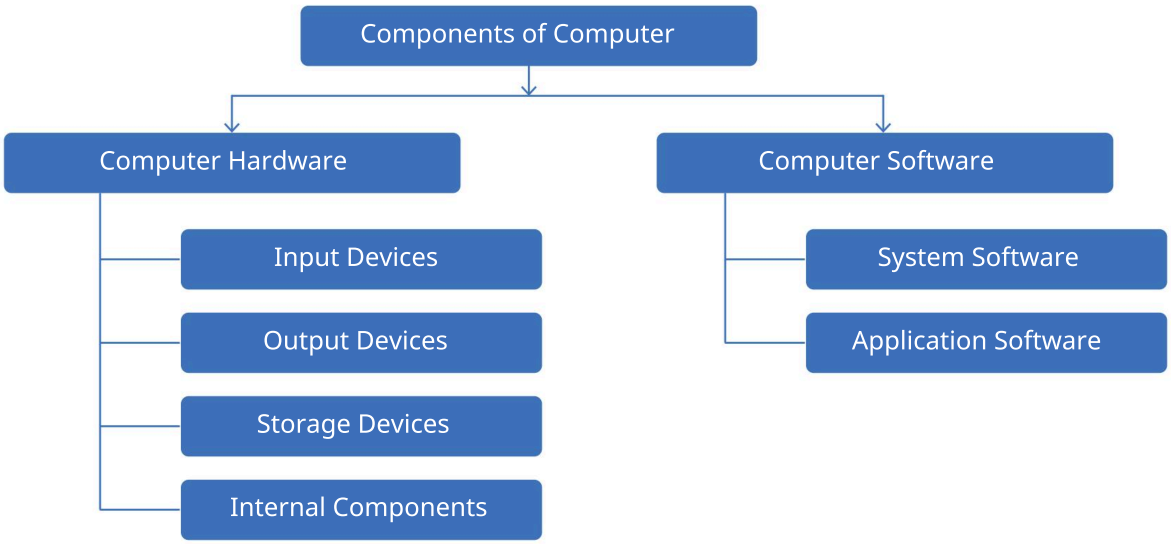
The motherboard is the primary circuit board that connects all the essential components of a computer system. It houses the CPU, memory, and the connectors for peripheral devices such as the hard drive, CD/DVD drive, and graphics card.

Storage Devices

Storage devices are used for long-term data storage. These can include Hard Disk Drives (HDDs), Solid-State Drives (SSDs), external storage devices like USB drives, and external hard drives.

Hardware and Software of a Computer System

Computer systems rely on both hardware and software components, which are integral for ensuring compatibility with the users.



Hardware

Hardware refers to the physical components of a computer system. These are physical devices that can be seen or touched. Computer hardware can be categorised into different types—input devices, output devices, storage devices, and internal components.

1. **Input Devices:** These devices enable the users to input data and communicate with the computer system. Some examples of input devices include the keyboard, a mouse, and a scanner.
2. **Output Devices:** These devices display the results of user-performed tasks. Some examples of output devices are monitors, printers, and speakers.
3. **Storage Devices:** These devices are used to store data and are often referred to as secondary storage. Some examples of storage devices include CDs, DVDs, and hard disks.
4. **Internal Components:** These critical hardware components are integral parts of the computer system. Some examples of internal components include the CPU and the motherboard.

Software

Software is a set of instructions or programs given to the computer to complete a task. It is a part of the computer that cannot be touched or felt. Some examples of software are Windows, OpenOffice Writer, Microsoft Excel, PowerPoint, Google Chrome, Photoshop, and MySQL.

There are two types of computer software—system software and application software:

1. **System Software:** System software controls the overall working of a computer. It manages all the input and output operations of the computer. For example, the operating system is a part of the system software that makes a computer run smoothly.
2. **Application Software:** Application software facilitates fundamental computer operations. It performs specific tasks for users. This category includes word processors, spreadsheets, and a variety of other task-specific programs. There are two types of application software: general-purpose software and customised software.

Did You Know?

The operating system manages computer hardware and software. It provides the users with an interface through which they communicate with the computer.

Role and Functions of RAM and ROM

In computer systems, the memory is a hardware component of the system that stores data and information. The computer memory can be classified into two main types—primary memory and secondary memory. The primary memory is further divided into two main types—RAM and ROM.

Random Access Memory (RAM)

RAM, or Random Access Memory, serves as the primary memory in a computer system. It temporarily holds data and instructions that the computer is currently processing. RAM allows the CPU to access data quickly, enabling efficient multitasking and the smooth execution of programs.

Since it is a volatile memory, RAM loses its data when the computer is powered off. RAM, also referred to as the main memory of the computer, enables the CPU to have direct access to all its memory cells. RAM is mainly composed of semiconductive materials and typically takes the form of integrated circuits (ICs).

Read Only Memory (ROM)

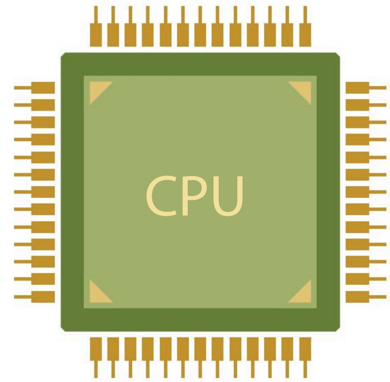
ROM, or Read Only Memory, constitutes a primary computer memory that is used to store instructions and programs that do not require any changes, such as the basic input/output system (BIOS). This storage aids in creating computer firmware, and data is generally stored during the manufacturing process. Similar to RAM, ROM is a type of semiconductor memory formed as integrated circuits (ICs).

Data stored in ROM is permanent, signifying that it is non-volatile and retains its data even when the computer is powered off. Unlike RAM, the CPU can only read data from the memory cells of ROM but is unable to modify it.

Difference Between RAM and ROM

Parameter	RAM	ROM
Full form	RAM stands for Random Access Memory.	ROM stands for Read Only Memory.
Definition	RAM is a primary memory of the computer that temporarily stores data and instructions on which the CPU is currently working.	ROM is a primary memory of the computer that stores the computer instructions and programs that do not need to be altered in future, like BIOS.
Nature	RAM is a volatile memory, which means it stores data as long as the computer system is turned on.	ROM is a non-volatile memory. Which means it stores data permanently even when the computer system is turned off.
Data access	The CPU of the computer can read, write, or alter the data on RAM.	The CPU can only read data from ROM, but it cannot write or change.
Types	There are two types of RAM: • SRAM (Static Random Access Memory). • DRAM (Dynamic Random Access Memory).	There are three types of ROM: • PROM (Programmable ROM) • EPROM (Erasable PROM) • EEPROM (Electrically EPROM).
Speed	The speed of RAM is quite high.	The speed of ROM is slower than RAM.
Cost	RAM is costly.	ROM is not so expensive.

Role and Functions of the Central Processing Unit (CPU)



The Central Processing Unit (CPU) interprets and executes instructions, performs tasks such as arithmetic operations, logic comparisons, and data movement. It is responsible for coordinating and managing the various components of the computer system, ensuring that instructions are carried out accurately and efficiently. The CPU has three main components, which are responsible for different functions: Arithmetic Logic Unit (ALU), Control Unit (CU), and Memory Unit (MU).

1. **The Arithmetic Logic Unit (ALU)**

   The Arithmetic and Logic Unit (ALU) is a crucial component of the CPU responsible for executing mathematical computations and logical decisions. It conducts basic arithmetic functions such as addition, subtraction, multiplication, and division, alongside logical comparisons that determine whether data items are larger, smaller, or equal. The ALU is essentially a foundational building block of the CPU, constituting a digital circuit designed specifically for carrying out arithmetic and logical operations.

2. **The Control Unit (CU)**

   The Control Unit (CU), a vital part of the computer's central processing unit, arranges and manages the flow of data to and from the CPU. It oversees the activities of the ALU, memory registers, and input/output units, and ensures the execution of all the instructions stored in the program. This unit decodes the fetched instructions, interprets them, and dispatches control signals to input/output devices, facilitating proper execution of operations by the ALU and the memory registers.

   The Control Unit serves as the director of the processor's activities, guiding the computer's memory, ALU, and input and output devices in responding to the processor's instructions.

3. **Memory Unit (MU)**

   A temporary memory unit within the CPU is in the form of memory registers. They serve the purpose of directly storing data utilised by the processor. They come in various sizes, such as 16-bit, 32-bit, 64-bit, and so forth. Each register in the CPU is designated for specific functions like data storage, instruction storage, and memory location addressing. Assembly language programmers can utilise user registers to store operands, intermediate results, and other essential data.

Did You Know?

The Accumulator (ACC), a pivotal register within the ALU, contains one of the operands required for the operation to be executed within the ALU.

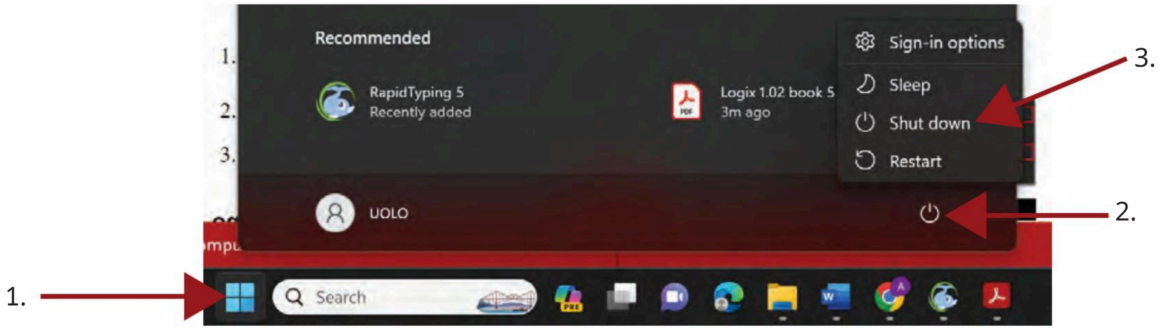
Procedure for Starting and Shutting Down a Computer

Starting Up

1. Connect the power cable and peripherals.
2. Press the **Power** button to start the computer.
3. Wait for the operating system to load.

Shutting Down

1. Save any open files and close running programs.
2. Click on the **Start** menu. Click on the **Power** button. Select **Shut down** option.
3. Wait for the computer to shut down completely before turning off the power.



Activity Time

Activity 1: Understanding Computer Parts ***(Group Work)***

Organise students into small groups and provide them with either computer setups or visual aids showcasing computer parts.

Activity 2: Recognising Computer Parts ***(Individual Work)***

Instruct students to recognise and name different hardware components, including the CPU, monitor, keyboard, mouse, and storage devices.

Activity 3: Group Discussion ***(Group Work)***

Create groups of 4–5 students and discuss the specific roles and contributions of each hardware element to the overall functioning of a computer system.

Chapter Checkup

A Select the correct option.

1. What is the primary function of the Control Unit in a CPU?
   - a Execute mathematical computations
   - b Manage data flow to and from the CPU
   - c Store temporary data within the processor
   - d Display results of user-performed tasks
2. .... is a type of application software.
   - a Device driver
   - b Operating system
   - c Word processor
   - d BIOS
3. What is the role of the motherboard in a computer system?
   - a Coordinates and manages the CPU
   - b Displays the results of user-performed tasks
   - c Connects all essential components
   - d Executes instructions and calculations

B Fill in the blanks with the most suitable words.

1. The CPU consists of three main components, namely the ...., Control Unit, and Memory Unit.
2. .... is a type of secondary memory used for long-term data storage.
3. The .... is responsible for executing instructions and performing calculations in a computer system.
4. .... devices enable users to input data and interact with the computer system.

C **State whether the following is *True* or *False*. Correct the statements that are false.**

1. The ALU performs arithmetic and logical operations within a CPU.
2. ROM is a volatile memory that loses data when the computer is powered off.
3. The motherboard houses the CPU, memory, and connectors for peripheral devices.
4. RAM is used for long-term data storage.

D **Answer the following questions.** ***(Solved)***

Q1. What are the two main types of memory in a computer system and how do they differ in their functions?

A1. The two main types of memory are RAM (Random Access Memory) and ROM (Read-Only Memory). RAM is used for temporary data storage, allowing the CPU to access data quickly for current operations. It is volatile, which means it loses its data when the computer is powered off. On the other hand, ROM is used to store essential software and firmware that cannot be altered. It is non-volatile, retaining its data even when the power is off. Unlike RAM, the CPU can only read data from ROM but cannot modify it.

Q2. Explain the role of the Control Unit in a computer's CPU.

A2. The Control Unit is a vital part of the CPU that manages and coordinates the flow of data to and from the CPU. It oversees the activities of the Arithmetic Logic Unit (ALU), memory registers, and input/output units, ensuring the execution of all instructions stored in the program. It decodes the fetched instructions, interprets them, and dispatches control signals to input/output devices, facilitating the proper execution of operations by the ALU and memory. It serves as the director of the processor's activities, guiding the computer's memory, ALU, and input and output devices in responding to the processor's instructions.

Q3. Akshit wants to know about the hardware of a computer system and its different categories. Explain it to him.

A3. Hardware refers to the physical components of a computer system. These are tangible, physical devices that can be seen or touched. Computer hardware can be categorised into different types, including input devices, output devices, storage devices, and internal components.

- **Input Devices:** These devices enable users to input data and interact with the computer system. Examples of input devices include the keyboard, mouse, and scanner.
- **Output Devices:** These devices display the results of user-performed tasks. Examples of output devices are monitors, printers, and speakers.
- **Storage Devices:** These devices are used to store data and are often referred to as secondary storage. Examples of storage devices include CDs, DVDs, and hard disks.
- **Internal Components:** These critical hardware components are integral parts of the computer system. Examples of internal components include the CPU and motherboard.

Answer Key

A **1.** b **2.** c **3.** c

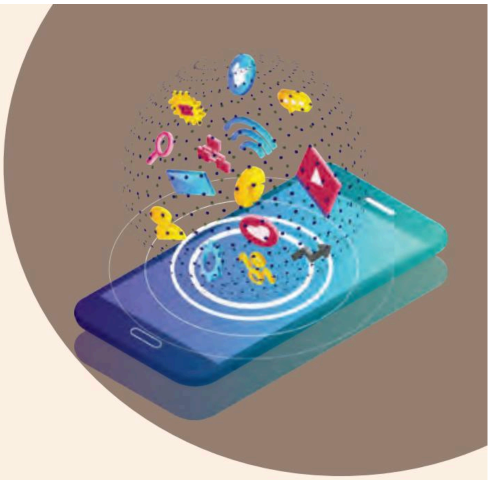
B **1.** Arithmetic Logic Unit (ALU) **2.** Hard Disk Drive (HDD) **3.** Central Processing Unit (CPU) **4.** Input

C **1.** True.

2. False. RAM is a volatile memory that loses data when the computer is powered off.

3. True.

4. False. ROM is used for long-term data storage.



9

Peripheral Devices

In this chapter, we will dive into the complex world of computer hardware, where we explore the various components and peripherals that make up a computer system. Understanding these elements is crucial, as they form the backbone of computing technology, enabling us to perform an array of tasks from simple calculations to complex simulations.

Have you ever thought about how your computer detects key presses and mouse movements? Or how your printer turns digital files into printed pages? This all is possible with the help of the special devices that connect to your computer, known as peripherals.

Peripheral Devices

Peripheral devices are the tools that expand the capabilities of computing systems. They serve as connectors between the digital and physical worlds, facilitating interaction, communication, and the exchange of information.

A peripheral device, whether inside or outside the computer, is a tool that links directly to a computer or digital device but isn't directly involved in its main function, like processing data. It helps users access and utilise the computer's features. While the computer can work without these devices, some, like the mouse, keyboard, or monitor, are crucial for user-computer interaction. These devices are also known as input–output (I/O) devices.

Importance of Peripheral Devices

Peripheral devices significantly contribute to improving our overall interaction with computing systems and unlocking the full capabilities of our devices. Let's explore why these tools are not merely add-ons but rather integral parts of our technological resources.

1. **Enhanced Functionality:** They expand the capabilities of computers and other digital devices, allowing users to perform a wider range of tasks, from simple operations to complex activities.
2. **User Interaction:** Peripherals such as a keyboard, mouse, and touchscreen enable users to interact with and control the digital system efficiently, making the user experience more intuitive and effective.
3. **Data Input and Output:** Devices like printers and scanners enable the conversion of digital data into physical formats and physical formats into digital data, respectively, making it easier to share and store information in various forms.
4. **Connectivity:** Peripherals such as modems, routers, and network adapters facilitate communication and data transfer between devices, enabling access to the internet and other networks.

5. **Efficiency and Productivity:** With the help of peripherals like external storage devices, users can store and retrieve data quickly, enhancing overall efficiency and productivity in various tasks.
6. **Specialised Tasks:** Certain peripherals are designed for specialised functions, such as graphic tablets for digital art, microphones for audio recording, and webcams for video conferencing, catering to specific user needs and preferences.

Types of Peripheral Devices

Let's study the various categories of peripheral devices and explore the wide range of functionalities that they offer.

Input Devices

Input devices are electronic devices that receive instructions from the external environment and convert that data into a format that the computer can read and interpret. They act as a crucial link between the outside world and the computer, facilitating effective communication. When users input data using these devices, the information can be stored in the computer's memory for later processing and manipulation. An input device transmits data to a computer, enabling users to interact with and manage it.

Types of Input Devices

The following are the various types of input devices:

Keyboard

The keyboard is the primary input device for feeding data into a computer. It comprises different keys for typing letters, numbers, and symbols. A keyboard closely resembles a typical typewriter. It also includes some keys for performing specific functions. Keyboards come in various varieties. The standard keyboard has 104 keys. It connects to a computer system using either a USB cable or a Bluetooth connection.

Characteristics of the Keyboard:

1. The keyboard has different keys to perform different functions.
2. It allows the use of arrow keys as an alternative to the mouse.
3. The main keys of a keyboard are the alphabet keys, special character keys, cursor keys, numeric keypad, and function keys.

Mouse

The mouse is the primary pointing device. Clicking and dragging the mouse enables the movement of the mouse pointer across the computer screen. The left mouse button facilitates item selection or relocation, while the right mouse button displays supplementary menus upon clicking. Some of the mouse types are trackball mouse, mechanical mouse, optical mouse, wireless mouse, etc.

Characteristics of the Mouse:

1. The mouse controls the movement of the cursor on the screen, enabling users to navigate in the desired direction.
2. It permits users to select files, folders, or multiple items or texts.
3. Users can hover the mouse pointer over any object.
4. The mouse can open files, folders, etc. This involves positioning the pointer over the file or folder and subsequently double-clicking to open or execute it.

Joystick

A joystick is a device used to navigate the cursor or pointer on a screen. It consists of a spherical ball attached to both ends of the stick, with the lower ball housed in a socket. The joystick allows for movement in all directions. They offer greater precision and durability compared to a mouse.

Characteristics of the Joystick:

1. It is used to control the position of the cursor or pointer on a screen.
2. It is commonly used for playing video games.
3. It has special push buttons that are pressed to provide input to the computer.

Light Pen

A light pen is a device resembling a pen that is used for pointing. It enables drawing on the monitor screen or selecting menu items. It contains a photocell and an optical system within a small tube. The photocell sensor component identifies the screen's position and transmits a signal to the CPU when the light pen's tip is moved across the monitor screen while the pen button is pressed.

Characteristics of the Light Pen:

1. The light pen is useful for drawing graphics.
2. It is used to select objects on the display screen.

Scanner

The scanner captures images, text, or other physical content from a physical source and converts them into a digital format that can be stored and manipulated on the computer. There are different types of scanners, like flatbed scanners, handheld scanners, sheetfed scanners, drum scanners, etc. Most scanners today are variations of the desktop (or flatbed) scanner. A flatbed scanner functions in a manner somewhat similar to a photocopier, with the key distinction being that a photocopier produces a physical paper copy, whereas a scanner creates a digital image that is stored on a computer.

Characteristics of the Scanner:

1. Scanners can handle low-quality or non-standard-weight paper.
2. They are versatile, enabling the scanning of various items regardless of their size. Whether small items or large documents, they can be scanned if appropriately positioned.

Webcam

A webcam is a video camera that captures and streams video in real time over the internet or a computer network. Webcams are commonly used for video conferencing, live streaming, online video chat, and video surveillance. They are often built into laptops, tablets, and smartphones, while external webcams can be connected to computers via USB or other interfaces.

Characteristics of a Webcam:

1. **Image Quality:** Webcams vary in image resolution and quality, typically measured in terms of pixels, with higher resolutions providing clearer and more detailed images.
2. **Frame Rate:** The frame rate represents the number of frames the webcam can capture per second, with higher frame rates leading to smoother and more fluid video.
3. **Connectivity:** Webcams can connect to computers through various interfaces, such as USB, Wi-Fi, or Bluetooth.
4. **Additional Features:** Some webcams come with built-in microphones, autofocus, low-light correction, and pan-tilt-zoom (PTZ) capabilities (this allows the camera to move horizontally, vertically, and zoom in and out), allowing for enhanced video and audio performance.
5. **Compatibility:** Webcams can be compatible with various operating systems and video conferencing software, although some may have specific compatibility requirements.

Barcode Reader

A barcode reader is a device that interprets information represented by light and dark lines, known as a barcode. A barcode is frequently used for price and product identification, labelling objects, numbering books, etc. The barcode reader is used to scan and extract data from barcodes. By shining light beams on the lines of the barcode, the reader identifies the encoded data within the barcode.

Characteristics of the Barcode Reader:

1. You can scan the barcode by simply positioning the barcode reader near the code and scanning it.
2. Visual indicators provide users with confirmation that the card has been swiped accurately.
3. Upon inserting a card, automated barcode scanners initiate scanning immediately.

Output Devices

An output device is any hardware component that receives information from a computer and then presents it in various forms, such as audio, visual displays, or printed copies. These devices are responsible for translating computer data into a format that can be easily comprehended by humans. While input devices facilitate data entry into the computer, it is the output devices that showcase the results of computer operations to the user.

Types of Output Devices

Let's learn about the various output devices.

Monitor

The primary output device of a computer is a monitor, commonly referred to as a visual display unit (VDU), which exhibits processed data, including text, images, videos, and audio. It achieves this by arranging minuscule dots, known as pixels. The clarity of the display, also known as its resolution, depends upon the number of pixels present. Higher resolution means more pixels are packed into the display area, hence better quality.

There are two types of monitors:

1. **Cathode-Ray Tube (CRT):** This monitor relies on a cathode-ray tube, which generates a stream of electrons via electron guns. These electrons strike the inner surface of a phosphorescent screen, creating images. The CRT monitor comprises millions of phosphorescent dots in three primary colours: red, blue, and green. These dots illuminate upon impact, resulting in image formation. Its key components include the electron gun, fluorescent screen, glass envelope, deflection plate assembly, and base.
2. **Flat Panel Monitor:** This type of display, in contrast to CRTs, is lighter, thinner, consumes less power, and has a higher resolution. It can be portable or mounted on walls, finding applications in devices such as calculators, video games, laptops, and graphical displays.

Did You Know?

Additionally, there is the plasma monitor, which is also a form of flat panel display. This technology utilises plasma cells positioned between two glass surfaces, containing noble gases and mercury solutions. When electricity is supplied, the gas transforms into plasma, generating UV light that produces an image.

Characteristics of the Monitor:

1. **Resolution Pixels:** Pixels are the smallest element of any image. The higher the number of pixels, the better the resolution of a monitor.
2. **Size:** The size of the monitor is the diagonal measurement of a desktop screen, which is typically 14 to 25 inches.
3. **Refresh Rate:** Total number of times per second that an image on a display is repainted or refreshed. The higher the refresh rate the better the display of the monitor.
4. **Luminance:** It is the brightness of the screen.

Printer

Printers are the output devices that produce a physical copy of a digital document or image. Printers are one of the most popular computer peripherals to print text and photos.

They are broadly classified into two categories:

1. **Impact Printers:** These types of printers employ a print head or hammer to transfer data onto the paper. The print head or hammer strikes an ink ribbon against the paper, resulting in the printing of characters. Some of the types of impact printers are:
   - Dot matrix printer
   - Daisy wheel printer
   - Line printer
   - Chain printer

2. **Non-impact Printers:** Non-impact printers are distinct in that they do not require a ribbon to print characters. These printers are frequently referred to as page printers due to their ability to print an entire page at once. Some of the types of non-impact printers are:
   - Laser printers
   - Inkjet printers

Characteristics of the Printer:

1. **Print Quality:** Printers vary in their ability to produce high-resolution text and images.
2. **Printing Speed:** The speed at which a printer can produce printed output varies, with some printers capable of high-speed printing for large volumes. The speed of a printer is measured in **pages per minute (ppm)**.
3. **Connectivity and Compatibility:** Printers may offer various connectivity options, such as USB, Wi-Fi, or Bluetooth, and compatibility with different operating systems and devices.

Plotter

A plotter is an advanced device used for producing high-resolution graphics in various colour formats. While similar to a printer in some aspects, it offers more sophisticated capabilities. Its primary applications include printing large-scale maps, architectural designs, and oversized graphics. Additionally, it is employed for creating images, 3D renderings, advertising materials, and detailed schematics of internal machine structures.

Characteristics of the Plotter:

1. Large-size prints can be taken via plotters.
2. It is slow and expensive.

Projector

A projector is a device that enables individuals to display their computer or device output on a wall or screen. Through the use of light and lenses, it enlarges and projects text, images, and videos. Consequently, it serves as a valuable output device for delivering presentations or projecting movies.

Characteristics of a Projector:

1. They are portable and can be effortlessly connected and used to project an image on a wall by a single individual.
2. Projectors represent a highly economical solution for creating a large video display within the home.
3. Projectors are compact and can be easily mounted on a back shelf, bookcase, or ceiling, as they occupy no floor space.

Speaker

Speakers are essential peripherals linked to computers for audio output. To facilitate their function, sound cards are necessary. Available in diverse configurations, from basic two-speaker setups to more elaborate surround-sound systems with multiple channels, speakers come in various sizes and designs. They receive audio input from the computer's sound card and translate it into audible sound waves.

Characteristics of Speaker:

1. Speakers are available in a wide range of qualities and prices.
2. Nowadays most computer systems include speakers in the CPU cabinets.

Microphone

A microphone is an audio input device that lets you talk or record sounds on your computer. Using a microphone, often called a mic, you can chat with friends, record your voice, or even control your computer using voice commands.

Characteristics of a Microphone:

1. A microphone can capture sounds from the surroundings very clearly.
2. It is easy to use by simply plugging it into your computer.
3. The microphones can be used for a variety of tasks such as online meetings, chatting with friends, voice recording, and giving voice commands.
4. Microphones used with computers are available in small sizes, making them convenient for desktop or laptop use.

Did You Know?

Most laptops these days are equipped with microphones, by default. With desktop computers, you can use either a pair of microphones or a headset that is a combination of microphones and speakers.

Storage Devices

Storage devices are integral components in which a computer retains all its data, including files, programs, operating system, etc. These devices are like virtual shelves, enabling the systematic organisation and retrieval of digital assets. Without storage devices, the computer would be unable to retain files, programs, or even its fundamental instructions. Storage devices safeguard and facilitate access to your data whenever required.

Types of Storage Devices

Let us learn about some commonly used storage devices.

Hard Disk Drive (HDD)

A Hard Disk Drive (HDD) comprises a rotating disk (platter) coated with a magnetic substance and a read/write head that records and retrieves data on the disk's surface. The read/write head moves across the spinning disk to access different sections of the stored data. HDDs are commonly used to backup data.

Characteristics of a Hard Disk Drive (HDD):

1. **Ample storage capacity:** HDDs provide substantial storage space, with certain models capable of accommodating up to 16TB of data.
2. **Cost-effectiveness:** HDDs offer a budget-friendly solution for storing extensive volumes of data.
3. **Larger physical dimensions**: Although HDDs are often installed inside the CPU cabinets, nowadays external hard disks are also available. These can easily be carried from one place to another.
4. **Slower operational speed**: HDDs exhibit slower data access and transfer speeds in comparison with primary memory.
5. **Mechanical components:** HDDs include mechanical elements that can deteriorate over time.

Solid State Drive (SSD)

Solid State Drives (SSDs) utilise flash memory for data storage rather than a spinning disk. SSDs have significantly faster speeds, enhanced durability, and reduced vulnerability to mechanical breakdowns compared to HDDs.

Characteristics of a Solid-state Drive (SSD):

1. **Faster performance:** SSDs deliver notably faster data access and transfer speeds in comparison to HDDs.
2. **Compact form:** With smaller and lighter physical dimensions, SSDs are well-suited for integration into portable devices like laptops and tablets.
3. **Lower power consumption:** SSDs consume less energy than HDDs, thereby promoting greater energy efficiency.
4. **Higher price point:** SSDs are more costly than HDDs.
5. **Absence of mechanical components:** SSDs lack moving parts, resulting in increased durability and decreased susceptibility to mechanical failure compared to HDDs.

Flash Drive

A flash drive, also known as a pen drive, is a portable storage device that utilises flash memory to store and transfer data. It is typically small in size and connects to a computer via a USB (Universal Serial Bus) port, allowing for easy data transfer between devices. Flash drives are commonly used for backing up data.

Characteristics of Flash Drives:

1. **Portability:** Flash drives are compact and lightweight, making them highly portable and convenient for on-the-go use.
2. **Plug-and-Play:** They can easily be connected to a computer's USB port without the need for additional software installation, enabling quick data access and transfer.
3. **High Compatibility:** Flash drives are compatible with various operating systems and devices, allowing for seamless data exchange between different platforms.
4. **Data Storage:** They offer varying storage capacities, ranging from a few gigabytes to several terabytes, providing ample space for storing and transferring different types of files.
5. **Reusability:** Flash drives can be erased and rewritten multiple times, allowing for the easy removal and updating of data as needed.

Communication Devices

Communication devices are instrumental in enabling the transmission of data and information between your computer and external networks, devices, or the internet. Their functionality empowers your computer to engage in data sharing, access resources, and establish connections with the global digital network. Whether it involves internet connectivity, video conferencing for collaborative purposes, or the smooth transfer of files, these devices are essential for enhancing the communication and interactive capabilities of your computer.

Types of Communication Devices

Let us learn about some commonly used communication devices.

Modem

A modem, short for modulator-demodulator, is a hardware device that allows a computer or other devices to connect to the internet. It modulates digital data into an analog signal that can be transmitted over telephone lines, cable lines, or fibre optics, and then demodulates the incoming analog signal back into digital data. There are several types of modems, including dial-up modems, DSL modems, cable modems, and fibre modems, each of which has its own specifications and capabilities.

Characteristics of the Modem:

The following are the characteristics of a modem:

1. **Connection Type:** Modems can connect through various mediums, including telephone lines (dial-up), coaxial cables (cable modems), DSL lines (DSL modems), or fibre optics (fibre modems).
2. **Speed:** Modems have different data transmission speeds, typically measured in bits per second (bps) or megabits per second (Mbps).
3. **Functionality:** Some modems may also include additional features such as built-in Wi-Fi routers, multiple ethernet ports, or advanced security features.

Router

A router is a device that helps different types of computer networks to communicate with each other. A modem connects a computer to the internet, whereas a router connects two computer networks to each other as well as the internet. The main functionality of a router is to make sure that the information reaches its intended destination while keeping data safe along the way.

Characteristics of a Router:

1. **Directing Traffic:** Routers find the best way for information to travel from one computer to another on a network.
2. **Security Features:** Routers can implement security features such as firewalls to protect the computer networks from any unwanted traffic.
3. **Wi-Fi:** Some routers let you connect to the internet without using cables, which is called Wi-Fi (Wireless-Fidelity).
4. **Updates:** Routers can be updated to get new features or fix problems.

Activity Time

Activity 1: Group Discussion *(Group work)*

Create groups of 4–5 students and ask them to discuss various input and output devices that they use in their daily tasks.

Activity 2: Research Work *(Individual work)*

Visit your nearby stores, such as a grocery shop, medical store, and a restaurant. Identify and list the peripheral devices that are used at these places. Compare your list with your classmates' and discuss.

Activity 3: Create a Collage *(Individual work)*

Collect the pictures of various peripheral devices from the old newspapers, books, or magazines. Create a collage by pasting these pictures on chart paper. Paste the collage in your classroom or computer lab.

Chapter Checkup

A Select the correct option.

1. Which of the following is a type of printer that does not require a ribbon to print characters?
   - a. Dot matrix printer
   - b. Line printer
   - c. Laser printer
   - d. Daisy wheel printer

2. What is the primary function of a modem?
   a. To print documents
   b. To project images
   c. To connect to the internet
   d. To scan barcodes

3. Which of the following is a characteristic of a flash drive?
   a. Large size and heavyweight
   b. Incompatibility with various operating systems
   c. Slow data transfer rate
   d. Portability and plug-and-play functionality

B. Fill in the blanks with the most suitable words.

1. .... is a device that enables individuals to display their computer or device output on a wall or screen.
2. .... devices receive information or instructions from the external environment and convert that data into a format that the computer can read and interpret.
3. .... devices are instrumental in enabling the transmission of data between your computer and external networks, devices, or the internet.

C. State whether the following is *True* or *False*. Correct the statements that are false.

1. A light pen is used primarily for drawing graphics and selecting objects on the display screen.
2. Solid-state drives (SSDs) are generally more affordable than hard disk drives (HDDs).
3. The monitor is the primary output device of a computer, displaying processed data including text, images, and audio.

D. Answer the following questions. *(Solved)*

Q1. What are the main differences between a solid-state drive (SSD) and a hard disk drive (HDD)?

A1. The main differences between a solid-state drive (SSD) and a hard disk drive (HDD) are:
- **Memory:** SSDs use flash memory, while HDDs have spinning disks.
- **Speed:** SSDs are faster.
- **Durability:** SSDs have no moving parts, making them more durable.
- **Power consumption:** SSDs consume less power.
- **Cost:** SSDs are generally more expensive than HDDs.

Q2. How do input devices facilitate effective communication between the external environment and the computer? Provide examples of commonly used input devices.

A2. Input devices facilitate effective communication by receiving information or instructions from the external environment and converting that data into a format that the computer can read and interpret. Examples of commonly used input devices include the keyboard, mouse, scanner, and barcode reader.

Q3. In Samita's class, the teacher is using the projector to display a presentation on peripheral devices. Explain some characteristics of projectors.

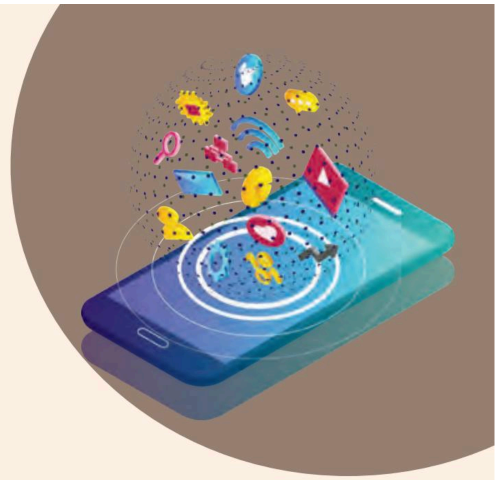
A3. The following are the characteristics of projectors:
- They are portable and can be effortlessly connected and used to project an image on a wall by a single individual.
- Projectors represent a highly economical solution for creating a large video display within the home.
- Projectors are compact and can be easily mounted on a back shelf, bookcase, or ceiling, as they occupy no floor space.

Answer Key

A 1. c 2. c 3. d

B 1. Projector 2. Input 3. Communication

C 1. True.
2. False. SSDs are generally more expensive than HDDs.
3. True.



10

Basic Computer Skills

You have learnt about various peripheral devices in the previous chapters that are used for providing input to the computer, store data, and give output. Let us now learn about some basic computer skills that are required to perform operations on the computer in this chapter.

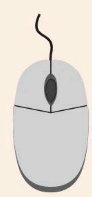
Primary Operations

Computer systems perform several primary operations, which include input, output, storage, processing, and control. Input involves providing data to the system; processing involves performing calculations on the data; output involves providing data to the user; and storage involves saving data. Together, these operations enable the system to function properly and perform various tasks. Let us learn about these one by one.

Input Operations

The process of supplying data to a computer system for processing is known as input operation. This can be done through various devices, such as a keyboard, mouse, microphone, web camera, etc. The data is then translated into a format that the computer can understand and process. Input is crucial to the functioning of a computer system, as it allows the users to interact with the system and provide it with the necessary information to perform various tasks.

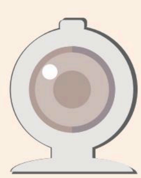
Keyboard



Mouse

Joystick

Scanner



Web Camera

Microphone

Input Devices of Computers

There are several types of input devices in a computer system. The most common ones are:

Keyboard Typing on the keyboard provides information to the computer. Keyboards come in a variety of sizes and layouts, including standard QWERTY keyboards and ergonomic keyboards designed to reduce hand and wrist strain.

Mouse A mouse is used to point and click on the screen. It comes in a variety of shapes and sizes, including wired and wireless models, and can be customised with additional buttons and features.

Scanner It is a device that scans an image (such as photographs, printed text, or handwriting) or an object (such as an ornament) and converts it to a digital image. Most scanners today are variations of the desktop (or flatbed) scanner. The flatbed scanner is commonly used in offices. A flatbed scanner functions in a manner somewhat similar to a photostat machine, with the key distinction being that a photocopier produces a physical paper copy, whereas a scanner creates a digital image that is stored on a computer.

Microphone A microphone, sometimes referred to as a mike or mic, is a device that converts sound into an electrical signal. They are used in many applications, such as telephones, hearing aids, recording studios, etc. All microphones capture sound waves with a thin, flexible diaphragm. The vibrations of this element are then converted by various methods into an electrical signal, which is an analog of the original sound.

Web Camera It is used to capture visual information and feed it into the computer. This information can be processed and used by the computer in a variety of ways, commonly in video conferencing.

Joystick A joystick is a pointing device used commonly for playing games. It includes a little handle that you can move in different directions. When you push or pull it, your game character does the same. Some joysticks also have buttons that you can press to make your character jump or do some other action.

Did You Know?

The touchscreen is both an input and output device. It is used on devices like smartphones and tablets where users interact with the screen directly. Users can swipe, tap, and pinch to zoom, providing input to the computer naturally and intuitively. The display can show visual outputs too.

Processing Operations

Processing operations are essential for the functioning of a computer system and involve manipulating data and instructions by the CPU. These operations include arithmetic, logical, input/output, control, and data movement operations. They control the flow of data and instructions, transfer data between the computer's memory and external devices, and perform basic mathematical functions. The speed and efficiency of processing operations depend on factors such as the CPU's clock speed, cache size, and number of processing units.

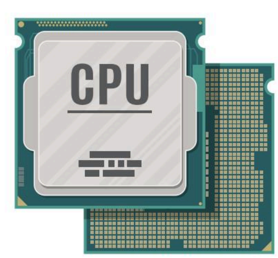
Processing Devices of a Computer

There are several types of processing devices in a computer system. The most common ones are:

Central Processing Unit (CPU) The CPU is the brain of the computer, responsible for executing instructions and managing the system's resources.

Graphics Processing Unit (GPU) The GPU is a specialised processor designed to handle complex calculations related to graphics and video processing.

Digital Signal Processor (DSP) A DSP is a microprocessor optimised for processing digital signals, such as audio and video.



Central Processing Unit (CPU)/Processor

Application-specific Integrated Circuit (ASIC) An ASIC is a specialised chip designed for a specific application or task, such as encryption or data compression.

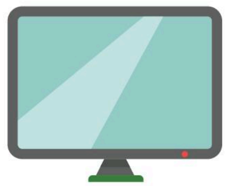
Field-programmable Gate Array (FPGA) An FPGA is a programmable chip that can be configured to perform a wide range of functions, making it useful for applications that require flexibility and adaptability.

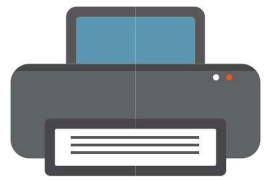
Output Operations

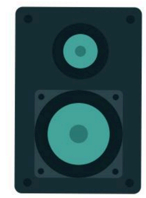
The process of displaying or producing information in a computer system is known as output operations. The computer produces output in various formats, such as text, graphics, sound, and video. Output is just as important as input because it allows users to see and perceive the information that the computer has processed.

Output Devices of a Computer

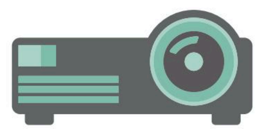
There are several types of output devices in a computer system. The most common ones are:


Monitor


Printer


Speaker


Headphones


Projector

Printers They are used to present text, images, and other information in a physical form, making it easier to read and share. Printing can be performed using a variety of printing technologies, including inkjet, laser, and dot matrix printers.

Monitors They are used to display text, images, and videos to the user. They are available in a wide range of sizes and resolutions, from small handheld devices to large wall-mounted displays.

Speakers Speakers serve the purpose of delivering sound to the user, including music, sound effects, and voice recordings. Audio can be played through built-in speakers on a computer or through external speakers or headphones.

Projectors They are used to project visual information onto a larger screen or surface. They are commonly used in presentations, movie theatres, and events and come in various types, including LCD projectors and DLP projectors.

Headphones They serve the purpose of delivering sound directly to the user's ears. They are commonly used for listening to music, watching videos, and playing games. They come in various types, including earbuds, over-ear headphones, and noise-cancelling headphones.

Plotter A plotter is a vector graphics printing device that connects to a computer. Plotters print their output by moving a pen across the surface of a piece of paper. This means that plotters are restricted to line art rather than raster graphics, as with other printers. Because of the mechanical movement of the pens, they can draw complex line art, including text, although they do it slowly.

Storage Operations

Storage operations are crucial for saving and retrieving data. They involve reading and writing data to and from these storage devices. This is done through input/output operations, which involve transferring data between the storage device and the computer's memory or processor. There are two main types of storage devices: primary storage and secondary storage.

Primary Storage

It is temporarily used to hold data that is currently being processed by the computer. It includes:

Random Access Memory (RAM) The RAM chip provides volatile storage, where programs and data are temporarily saved during processing and permanently deleted when the computer is turned off. RAM allows the CPU to quickly read and write data, which significantly speeds up the computer's performance.

Read-only Memory (ROM) The ROM chip retains the stored programs and data even when the computer turns off. So, unlike RAM chips, ROM chips are non-volatile. ROM contains data that is read-only, meaning it can be read but cannot be easily modified by the user. The ROM chip contains microprogrammed control instructions that instruct the machine to perform certain operations. For example, the BIOS (Basic Input/Output System) in a computer is typically stored in the ROM, ensuring that the computer can boot up and initialise its hardware components.

Cache Memory It is a high-speed memory. It can be inserted either on a motherboard or as a part of the CPU. It stores information and various instructions that are used repeatedly to execute the programs.
It improves the overall performance of the computer system in terms of the execution of commands. Cache is more expensive than RAM, but the user can also buy a CPU with in-built cache memory. The CPU first examines the cache when it wants certain data.

Secondary Storage

It is used for the long-term storage of data. It includes devices such as hard disk drives, solid-state drives, CDs, pen drives, memory cards, etc. These devices are used to store files and programs that are not currently being used by the computer.

Storage Devices of Computers

There are several types of storage devices in a computer system. The most common ones are:

Hard Disk Drive (HDD) HDDs are known for their large capacity, making them ideal for storing large files and programs. Hard disks are made up of one or more magnetic disks called platters. These platters store data and spin rapidly while a read/write head moves across them to access or modify data.


Hard Disk Drive

Solid-state Drives (SSDs) They are a newer type of storage device that uses flash memory to store data. SSDs are faster and more reliable than HDDs, making them a popular choice for gamers, video editors, and other users who require high-speed access to their data.

Solid-state Drive

Flash Drives They are also known as pen drives. They are small, portable storage devices that can be easily connected to a computer's USB port. They are ideal for transferring files between computers or for storing important data that can be easily carried around.


Flash Drive

Memory Cards They are small, removable storage devices that are commonly used in cameras and other portable devices. They are available in a range of sizes and capacities, making them a flexible storage option for users who need to store and transfer data on the go.


Memory Card

External Hard Drives They are larger storage devices that connect to a computer via USB or another interface. They are ideal for users who need to store large amounts of data or who require a backup solution for their important files and programs.


External Hard Drive

Communication Networking of Computer Systems

Communication networking in computer systems refers to the exchange of data and information between two or more computers or devices. Communication networking allows computers and devices to share resources, communicate with each other, and work together regardless of their physical location.

Here are some common types of computer networks used in communication networking:

Local Area Network (LAN) A LAN is a network that connects computers and devices within a limited geographical area, such as an office, building, or campus.

Metropolitan Area Network (MAN) A MAN typically spans a city or a large campus, connecting multiple LANs within that specific geographic area.

Wide Area Network (WAN) A WAN is a network that connects computers and devices over a large geographical area, country, continent, or even the world, using technologies such as the internet.

Wireless Network A wireless network uses wireless communication technologies, such as Wi-Fi, to connect computers and devices to the network.

Bluetooth Network A Bluetooth network allows devices to communicate wirelessly over short distances, typically within a range of 30 feet.

Virtual Private Network (VPN) A VPN is a network that allows users to connect to a private network over the internet. VPNs are commonly used for secure remote access to corporate networks.

Activity Time

Activity 1: Group Discussion *(Group Work)*

Make a group of 3 or 4 students and ask them to discuss input and output operations in class.

Activity 2: Creating a Chart *(Individual Work)*

Tell every student to make a chart of their favourite operation of a computer system.

Activity 3: Research Work *(Individual Work)*

Ask the students to conduct a research in their neighbourhood and find out about the various types of computer networks that these places use to communicate with each other.

Chapter Checkup

A Select the correct option.

1. Which of the following is NOT a primary operation of a computer?
   a. Input
   b. Processing
   c. Storage
   d. Printing
2. Which of the following is an example of an input device?
   a. Monitor
   b. Printer
   c. Keyboard
   d. Speaker
3. Which of the following refers to the saving and retrieval of data on a computer?
   a. Input
   b. Output
   c. Storage
   d. Control

B Fill in the blanks with the most suitable words.

1. The primary operations of a computer include input, processing, storage, and ....
2. A joystick is an example of an .... device.
3. .... is a type of storage device that uses flash memory to store data.
4. A .... is a network that connects computers and devices around the globe.

C State whether the following is *True* or *False*. Correct the statements that are false.

1. Processing refers to the manipulation of data by a computer.
2. A computer network does not allow computers and devices to share resources.
3. Input is the process of entering data and commands into a computer.
4. A webcam is an example of an input device.

D Answer the following questions. *(Solved)*

Q1. What are storage operations in a computer system?

A1. Storage operations are crucial for saving and retrieving data. It involves reading and writing data to and from these storage devices.

Q2. What is communication networking?

A2. Communication networking in computer systems refers to the exchange of data and information between two or more computers or devices. Communication networking allows computers and devices to share resources, communicate with each other, and work together regardless of their physical location.

Q3. Riya wants to backup her files to protect her data from potential loss. Name one device that she can use for this purpose.

A3. External hard drive.

Answer Key

A 1. d 2. c 3. c

B 1. Output 2. Input 3. SSD 4. WAN

C 1. True.

2. False. A computer network allows computers and devices to share resources.
3. True.
4. True.

Unit Reflection

Key Terms

ICT: ICT stands for Information and Communication Technology. It is a broad term that includes a wide range of technologies and tools used for handling, processing, storing, and communicating information.

Peripheral devices: Peripheral devices are the tools that expand the capabilities of computing systems. They serve as connectors between the digital and physical worlds, facilitating interaction, communication, and the exchange of information.

Global Positioning System (GPS): GPS is a radio-based satellite navigation system that uses radio signals to precisely determine a specific position.

Solid State Drives (SSDs): SSDs are storage devices that utilise flash memory for data storage rather than a spinning disk. SSDs have significantly faster speeds, enhanced durability, and reduced vulnerability to mechanical breakdowns compared to hard-disk drives (HDDs).

Pixels: Pixels are the smallest element of any image.

Cache: It is a high-speed memory. It can be inserted either on a motherboard or as a part of the CPU. It stores various information and instructions that are used repeatedly to execute the programs.

The Arithmetic and Logic Unit (ALU): It is a crucial component of the CPU responsible for executing mathematical computations and logical decisions.

Motherboard: It is the primary circuit board that connects all the essential components of a computer system. It houses the CPU, memory, and connectors for peripheral devices such as the hard drive, CD/DVD drive, and graphics card.

Things to Remember

- ICT has deeply impacted our personal lives. It has made convenience, efficiency, connectivity, and accessibility of information and resources at every doorstep possible.
- Key components of ICT are computers, software, the internet, telecommunications, networking, data storage, information security, and multimedia.
- ICT tools like smartphones, tablets, radio, televisions, laptops, and computers are part and parcel of modern society.
- The clarity of the display, also known as resolution, depends upon the number of pixels present. Higher resolution means more pixels are packed into the display area.
- Hardware refers to the physical components of a computer system. These are physical devices that can be seen or touched.
- A barcode reader is a tool that interprets information represented by light and dark lines in barcoded data. Barcoded data is frequently used for price and product identification, labelling objects, numbering books, etc.
- Software is a set of instructions or programs given to the computer to complete a task. It is a part of a computer that cannot be touched or felt.

- The Central Processing Unit (CPU) interprets and executes instructions and performs tasks such as arithmetic operations, logic comparisons, and data movement.
- The Control Unit, a vital part of the computer's central processing unit, arranges and manages the flow of data to and from the CPU.
- Tablets are portable computing devices that are bigger than smartphones but smaller than laptops.
- An e-newspaper, or electronic newspaper, is a digital version of a traditional print newspaper that is accessible through electronic devices such as computers, tablets, smartphones, and e-readers.
- Registers, a type of temporary memory unit within the CPU, serve the purpose of directly storing data utilised by the processor.
- Peripheral devices are the tools that expand the capabilities of computing systems. They serve as connectors between the digital and physical worlds, facilitating interaction, communication, and the exchange of information.
- Input devices are electronic devices that receive instructions from the external environment and convert that data into a format that the computer can read and interpret.
- An output device is any hardware component that receives information from a computer and then presents it in various forms, such as audio, visual displays, or printed copies.
- Storage devices are integral components in which a computer retains all its data, including files, programs, operating system, etc.
- Communication devices are instrumental in enabling the transmission of data and information between your computer and external networks, devices, or the internet.
- The process of supplying data to a computer system for processing is known as input operation. This can be done through various devices, such as a keyboard, mouse, microphone, web camera, etc.
- Processing operations are essential for the functioning of a computer system and involve manipulating data and instructions by the CPU.
- Storage operations are crucial for saving and retrieving data. They involve reading and writing data to and from these storage devices.
- Communication networking in computer systems refers to the exchange of data and information between two or more computers or devices.

Test Your Knowledge

A. Select the correct option.

1. Smartphones have .... for maps.
   - a. GPS
   - b. PGS
   - c. SGP
   - d. GSP
2. .... requires an internet connection and allows users to attach files in the form of images, audio, video, or documents.
   - a. E-readers
   - b. E-newspaper
   - c. Radio
   - d. Email
3. What type of device is a CRT?
   - a. Output
   - b. Input
   - c. Storage
   - d. Processing
4. .... is a device that scans an image (such as photographs, printed texts, or handwritings) or an object (such as an ornament) and converts it to a digital image.
   - a. Barcode reader
   - b. Scanner
   - c. Printer
   - d. Microphone
5. .... is the primary memory of the computer that stores data and instructions on which the CPU is currently working.
   - a. RAM
   - b. ROM
   - c. Cache memory
   - d. None of these

B. Fill in the blanks with the most suitable words.

1. .... is called the brightness of the screen.
2. ICT stands for ....
3. A .... is a device that enables individuals to display their computer or device output on a wall or screen.
4. A .... is a device that converts sound into an electrical signal.
5. .... software includes word processors, spreadsheets, and a variety of other task-specific programs.

C. State whether the following is *True* or *False*. Correct the statements that are false.

1. Google Drive provides a cloud storage service.
2. A light pen enables drawing on the monitor screen or selecting menu items.
3. The motherboard is the primary circuit board that connects all the essential components of a computer system.
4. The CU is known as the brain of the computer.
5. LANs connect computers across larger geographical areas.

D. Short answer-type questions.

1. What is the use of ICT in banking and finance?
2. What are SD cards?
3. Name the three types of ROM.

E. Long answer-type questions.

1. What is a modem? What are its various types?
2. What is the role of ICT in personal life?
3. Differentiate between system software and application software.

F. Competency-based questions.

1. Rani got a new computer. But she does not know how to start it. Help her by telling her the procedure to start her computer.
2. It was Maira's first day at school. During a lecture in the computer lab, she noticed that her teacher was using a device resembling a pen to draw on the monitor screen. Name the device.